

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Report Analysis

Questions

1.A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system.

Aim

To perform a detailed security analysis of the target system (**192.168.1.10**) using the Nmap tool, identify the open ports and services running on the target machine, analyze the possible security vulnerabilities associated with each service, understand how attackers exploit these vulnerabilities, and recommend suitable security mechanisms to secure the network infrastructure.

Objectives

- Identify open ports and network services.
- Analyze the security risks associated with each service.
- Understand how attackers exploit vulnerable services.
- Identify operating system information.
- Recommend security best practices.
- Understand the importance of network monitoring.

Tools Required

- Kali Linux
- Nmap
- Terminal
- Virtual Machine (VMware/VirtualBox)

Commands Used

Basic Scan

Bash

```
nmap 192.168.1.10
```

This command scans the target host and displays all open ports and active services.

Service Version Detection

Bash

```
nmap -sV 192.168.1.10
```

This command identifies the version of each service running on the target system.

Operating System Detection

Bash

```
nmap -O 192.168.1.10
```

This command attempts to determine the operating system installed on the target machine.

Aggressive Scan

Bash

```
nmap -A 192.168.1.10
```

This command performs advanced scanning, including operating system detection, version detection, script scanning, and traceroute.

Vulnerability Scan

Bash

```
nmap --script vuln 192.168.1.10
```

This command executes vulnerability detection scripts to identify known security issues.

```
kali@kali:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 lib-1.3.1 libpcap-1.10.5 nmap-libndnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:40 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0023s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5860/tcp  open  sip
5861/tcp  open  sip-tls
8880/tcp  closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 88.31 seconds

kali@kali:~$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:43 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0046s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
53/tcp    closed domain
80/tcp    open  http       Apache httpd 2.4.7
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5860/tcp  open  sip?
5861/tcp  open  sip-tls?
8880/tcp  closed http-proxy
8885/tcp  closed unknown
```

```
kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:49 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
5860/tcp  open  sip?
5861/tcp  open  sip-tls
8880/tcp  closed http-proxy
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.05 seconds

kali@kali:~$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:51 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0027s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http       Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5860/tcp  open  sip?
5861/tcp  open  sip-tls?
8880/tcp  closed http-proxy
Device type: bridgeVoIP adapter[general purpose]
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT0T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:adamy_gasparovskii:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT0T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.77 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 94.05 seconds

kali@kali:~$ sudo nmap -A scanme.nmap.org -oN Assignment1_Nmap_Report.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:54 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0028s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http       Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5860/tcp  open  sip?
5861/tcp  open  sip-tls?
8880/tcp  closed http-proxy
Device type: bridgeVoIP adapter[general purpose]
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT0T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:adamy_gasparovskii:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT Bridge (94%), AT0T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.40 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 93.15 seconds

kali@kali:~$ ls
172.25.83.194 Assignment1_Nmap_Report.txt Desktop Documents Downloads ethical ethicalhacking index.html index.html.1 Music Pictures practical Public sample1.txt sample.txt Templates Videos
```

```
kali@kali:~$ sudo nmap -A scanme.nmap.org -oN Assignment1_Nmap_Report.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:54 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0028s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
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143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5860/tcp  open  sip?
5861/tcp  open  sip-tls?
8880/tcp  closed http-proxy
Device type: bridgeVoIP adapter[general purpose]
Running (JUST GUESSING): Oracle Virtualbox (94%), Slirp (94%), AT0T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:adamy_gasparovskii:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT Bridge (94%), AT0T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.40 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 93.15 seconds

kali@kali:~$ ls
172.25.83.194 Assignment1_Nmap_Report.txt Desktop Documents Downloads ethical ethicalhacking index.html index.html.1 Music Pictures practical Public sample1.txt sample.txt Templates Videos
```

Additional Security Measures

- Configure firewalls.
- Enable Intrusion Detection Systems (IDS).
- Enable Intrusion Prevention Systems (IPS).
- Conduct vulnerability assessments.
- Monitor logs continuously.
- Perform regular security audits.
- Follow the principle of least privilege.

Conclusion

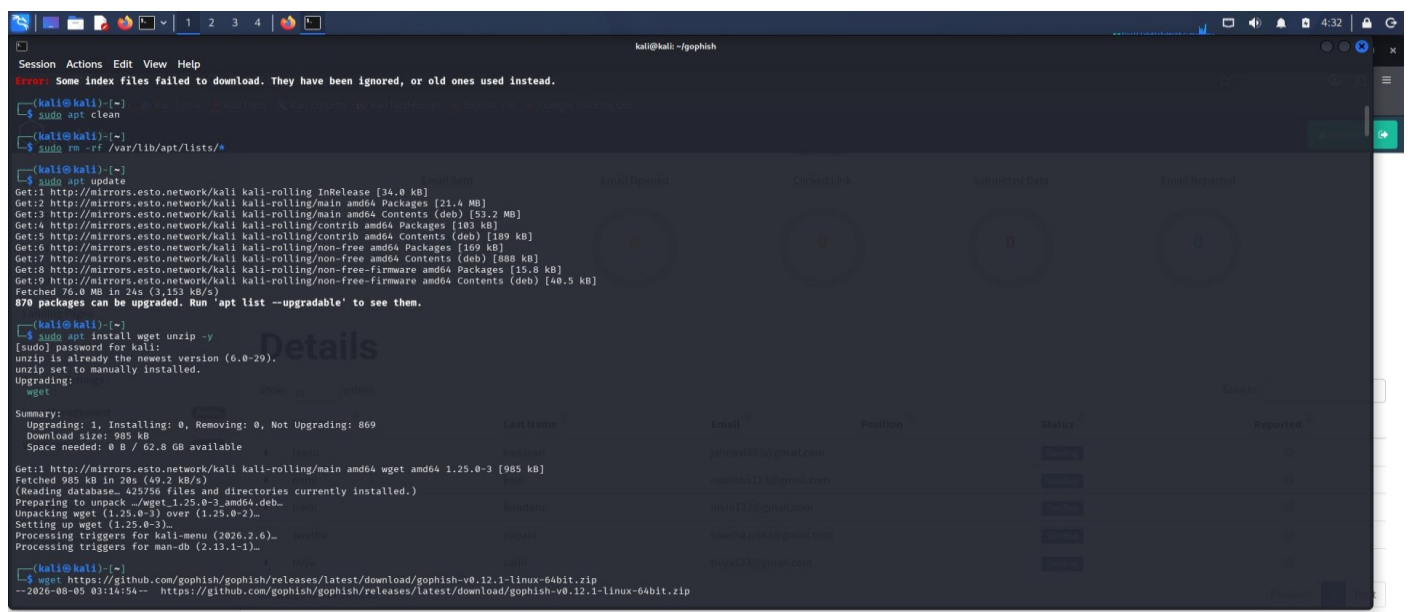
The Nmap scan identified several open ports and services that may expose the system to security threats. Proper configuration, strong authentication mechanisms, encryption, continuous monitoring, and timely software updates are essential to secure the network infrastructure and reduce cyber risks.

2.A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.

Aim

To analyze employee awareness regarding phishing attacks using GoPhish and identify security weaknesses that may lead to data breaches and unauthorized access.

Command 1 : sudo apt update



```
kali@kali: ~/gophish
Error: Some index files failed to download. They have been ignored, or old ones used instead.

(kali@kali)~$ sudo apt clean
(kali@kali)~$ sudo rm -rf /var/lib/apt/lists/*
(kali@kali)~$ sudo apt update
Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get:4 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [183 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]
Get:6 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [189 kB]
Get:7 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]
Get:8 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Packages [15.8 kB]
Get:9 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Contents (deb) [48.5 kB]
Fetched 76.0 MB in 24s (3,153 kB/s)
870 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)~$ sudo apt install wget unzip -y
[sudo] Password for kali:
unzip is already the newest version (6.0-29).
unzip set to manually installed.
Upgrading:
  wget
Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 869
  Download size: 985 kB
  Space needed: 0 B / 62.8 GB available

Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 wget amd64 1.25.0-3 [985 kB]
Fetched 985 kB in 20s (49.2 kB/s)
(Reading database... 42576 files and directories currently installed.)
Preparing to unpack ./wget_1.25.0-3_amd64.deb...
Unpacking wget (1.25.0-3) over (1.25.0-2)...
Setting up wget (1.25.0-3)...
Processing triggers for kali-menu (2026.2.6)...
Processing triggers for man-db (2.13.1-1)...

(kali@kali)~$ wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
--2026-08-05 03:14:54-- https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
```

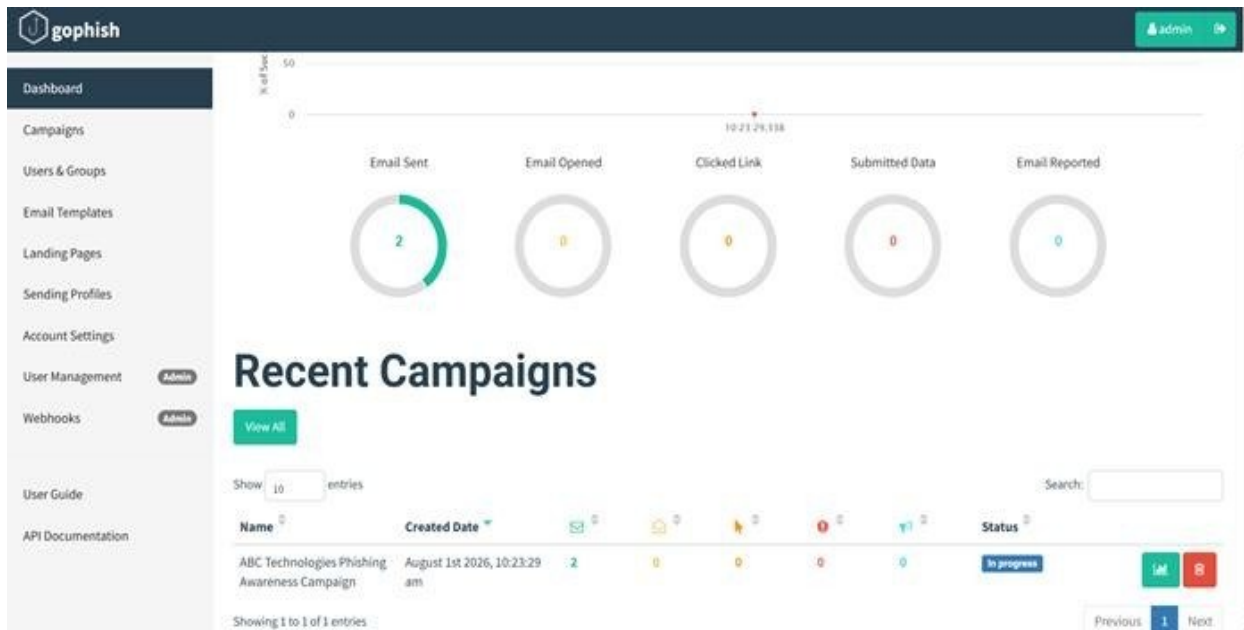
Package Name	Version	Architecture	Size	Status
wget	1.25.0-3	amd64	985 kB	Upgrading

Command 2 : `wget https://github.com/gophish/gophish/releases/latest/download/gophishv0.12.1-linux-64bit.zip`

Command 3 : `unzip gophish-v0.12.1-linux-64bit.zip -d gophish`

Command 4 : `cd gophish`

Campaign Results

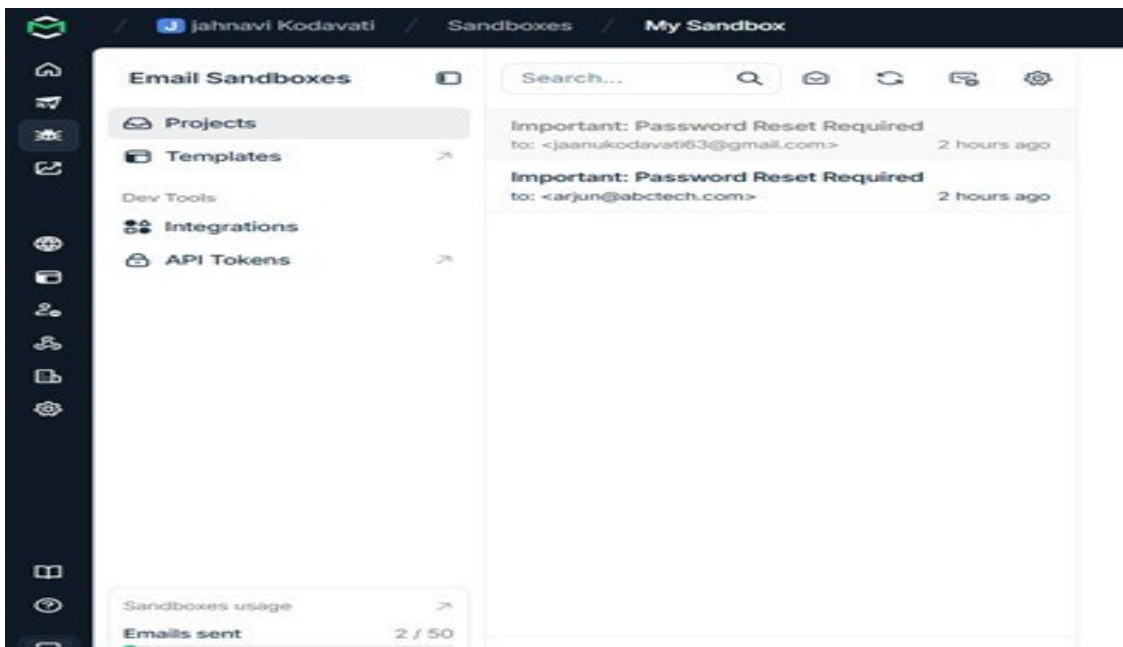
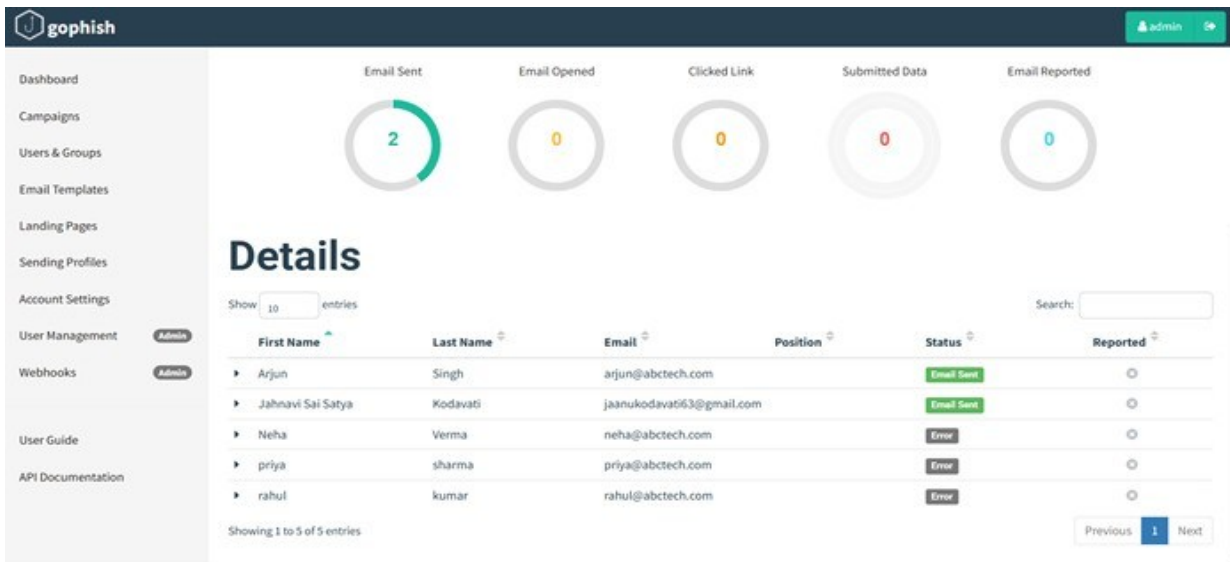


The screenshot shows the Gophish dashboard with the following metrics:

- Email Sent: 0
- Email Opened: 0
- Clicked Link: 0
- Submitted Data: 0
- Email Reported: 0

The dashboard also displays a table of campaign details:

First Name	Last Name	Email	Position	Status	Reported
jaanu	kodavali	jaanuv463@gmail.com		Sending	
namo	koyi	namtha123@gmail.com		Sending	
nishi	Bandaru	nishv123@gmail.com		Sending	
sawtha	papari	sawtha.p953@gmail.com		Sending	
thiya	sathi	thiya123@gmail.com		Sending	



Security Weaknesses

Lack of Awareness

Employees failed to identify fake emails and suspicious links.

Weak Password Practices

Employees shared sensitive credentials on fake websites.

Unsafe Attachment Downloads

Malicious files were downloaded without verification.

Social Engineering Attacks

Attackers exploited trust, fear, and urgency.

gophish

admin

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

User Guide

API Documentation

Email Sent

Email Opened

Clicked Link

Submitted Data

Email Reported

Details

Show 10 entries

Search:

First Name	Last Name	Email	Position	Status	Reported
jaanu	kodavati	jahnavi483@gmail.com		Sending	
nami	koyi	namitha123@gmail.com		Sending	
nikhi	Bandaru	nikhi123@gmail.com		Sending	
swetha	papani	swetha.p983@gmail.com		Sending	
tivya	sathi	tivya123@gmail.com		Sending	

Showing 1 to 5 of 5 entries

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gophish

admin

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

User Guide

API Documentation

Email Sent

Email Opened

Clicked Link

Submitted Data

Email Reported

Details

Show 10 entries

Search:

First Name	Last Name	Email	Position	Status	Reported
Arjun	Singh	arjun@abctech.com		Email Sent	
Jahnavi Sai Satya	Kodavati	jaanukodavati63@gmail.com		Email Sent	
Neha	Verma	neha@abctech.com		Error	
priya	sharma	priya@abctech.com		Error	
rahul	kumar	rahul@abctech.com		Error	

Showing 1 to 5 of 5 entries

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jahnavi Kodavati

Sandboxes

My Sandbox

Email Sandboxes

Projects

Templates

Dev Tools

Integrations

API Tokens

Sandboxes usage

Emails sent 2 / 50

Search...

Important: Password Reset Required to: <jaanukodavati63@gmail.com> 2 hours ago

Important: Password Reset Required to: <arjun@abctech.com> 2 hours ago

Security Recommendations

- Conduct monthly cybersecurity training.
- Implement phishing simulation exercises.
- Use spam filters and email gateways.
- Enable multi-factor authentication.
- Train employees to verify email addresses.
- Scan attachments before opening.

Conclusion:

The phishing awareness campaign revealed significant security weaknesses among employees, including a lack of awareness about suspicious links, credential theft attempts, and malicious attachments. The high percentage of users who clicked phishing links and shared credentials indicates the need for continuous cybersecurity training. Implementing strong email security controls and multi-factor authentication can significantly reduce phishing risks. Regular awareness programs and strict organizational security policies will help create a security-conscious workplace and minimize future social engineering attacks.

3.A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure.

Aim

To analyze DNS vulnerabilities and understand the security implications of DNS Zone Transfer attacks.

Commands Used

- dig AXFR
- zonetransfer.me @nsztml.digi.ninja

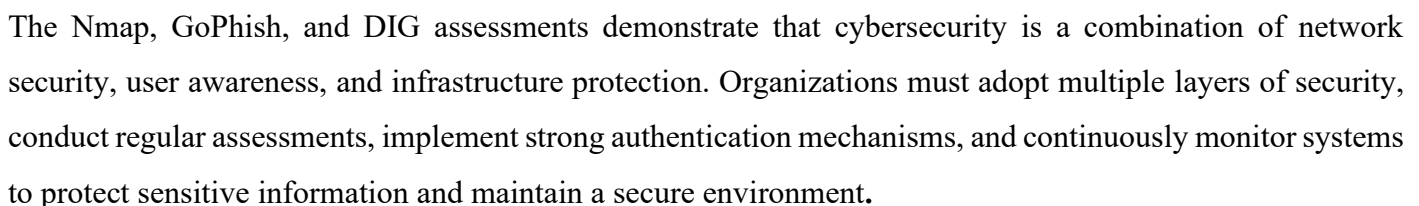
Security Risks

- Information leakage.
- Exposure of internal servers.
- Network reconnaissance.
- Phishing attacks.
- Increased attack surface.

Security Recommendations

- Disable public zone transfers.

- [illegible]



Code of Conduct:

I A.Sai Joshitha(192424220) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:A.Sai Joshitha

RUBRICS FOR CALCULATION

Report Analysis

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Report Analysis and Interpretation	25%	Accurately interprets the security report, identifies all key findings, and explains their significance with excellent technical understanding.	Correctly interprets most findings with minor omissions or inaccuracies.	Identifies only some findings and provides limited interpretation.	Unable to interpret the report or identify key findings.
Identification of Security Risks and Vulnerabilities	25%	Correctly identifies all security risks, vulnerabilities, and possible attack vectors with clear technical justification.	Identifies most risks and vulnerabilities with minor errors.	Identifies only a few risks with limited technical explanation.	Fails to identify major risks or vulnerabilities.
Application of Cybersecurity Concepts and Countermeasures	30%	Applies appropriate ethical hacking concepts and recommends comprehensive, practical, and technically sound countermeasures.	Recommends suitable countermeasures with minor technical gaps.	Provides limited or partially appropriate recommendations.	Recommendations are inappropriate, incomplete, or missing.

Technical Communication and Documentation	20%	Response is well-organized, technically accurate, clearly presented, and uses appropriate cybersecurity terminology throughout.	Response is organized and understandable with minor documentation or presentation issues.	Response lacks clarity, organization, or contains several technical inaccuracies.	Response is poorly organized, incomplete, and difficult to understand.
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